CS101 Introduction to Programming - Mid-Term Assessment (synthetic sample)
Time allowed: 90 minutes Total: 50 marks

Instructions: Answer ALL questions. Python 3 only; the standard library only.
Functions must RETURN their results (do not print or use input()).
Marks are awarded for correctness, handling of invalid input, and clear naming.

Question 1 (12 marks)
(a) Write a function grade_to_letter(score) that converts a score from 0 to 100 into a letter:
90-100 A, 80-89 B, 70-79 C, 60-69 D, below 60 F. Raise ValueError if score is outside 0-100.
(b) Write a function class_stats(scores) that takes a non-empty list of scores and returns a dict with the keys 'mean', 'median', 'highest' and 'lowest'. Raise ValueError if the list is empty.

Question 2 (14 marks)
Write a class BankAccount. It starts with a balance of 0 (or an optional opening balance). Provide deposit(amount) and withdraw(amount), both of which must reject non-positive amounts with ValueError. withdraw must raise a custom exception InsufficientFundsError (a subclass of Exception) if the balance is too low. Provide a balance property and a history() method returning a list of ('deposit'|'withdraw', amount) tuples in the order they happened.

Question 3 (10 marks)
A server log has lines such as: 2024-03-01 12:00:05 ERROR disk full
Write a function parse_log(lines) that returns a tuple (counts, errors), where counts is a dict mapping each level (INFO, WARNING, ERROR) to how many lines had it, and errors is the list of messages of the ERROR lines, in order. Ignore blank lines and lines that do not match the format.

Question 4 (6 marks)
Explain, in your own words and in no more than five sentences, the difference between a list and a tuple in Python, and give one situation where a tuple is the better choice.

Question 5 (8 marks)
Write a generator function first_primes(n) that yields the first n prime numbers, starting from 2.
For n = 0 it yields nothing. Raise ValueError if n is negative.